

HES FDA Toolkit — Usage Guide

Revision History

Date	Version	Author	Note
2026-08-12	1.0	Hau Pham	Initial version

Prerequisites

- **Python 3.x** installed:

```
python --version
```

- **WSL Ubuntu-22.04** installed (required for **WFDB 10.7.0** — the EC57 scoring library):

```
wsl --install -d Ubuntu-22.04
```

- Run **setup.bat** once:

```
setup.bat
```

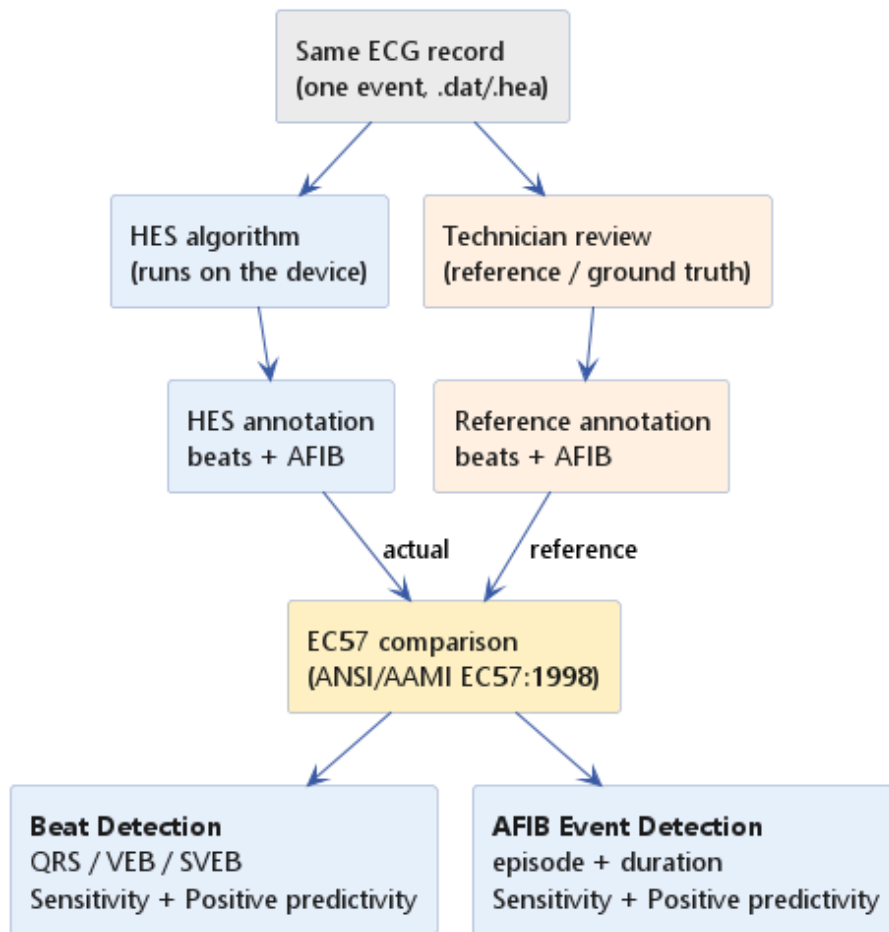
- **Launch the UI:**

```
run.bat
```

Overview

Evaluate the HES arrhythmia algorithm against ANSI/AAMI EC57:1998 (FDA-recommended) for Beat Detection and AFIB Event Detection, using technician annotations as the reference. The workflow is shown below:

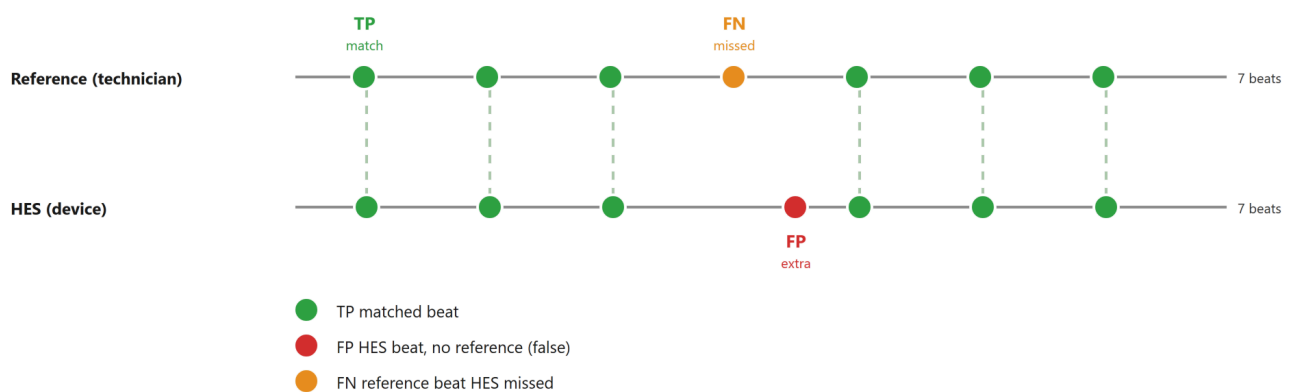
Performance evaluation of the HES algorithm — EC57:1998



Beat Detection

Each beat HES marks is matched, within a small time window, to the technician's beats:

- **True Positive (TP)** — an HES beat that matches a reference beat.
- **False Positive (FP)** — an HES beat with no reference beat (a false detection).
- **False Negative (FN)** — a reference beat that HES missed.



Sensitivity (Se)

$$TP / (TP + FN)$$

of the real beats, how many HES found

$$6 / (6 + 1) = 85.7\%$$

Positive predictivity (+P)

$$TP / (TP + FP)$$

of HES's beats, how many are real

$$6 / (6 + 1) = 85.7\%$$

Beats are graded by AAMI class:

- 1. **N**: Normal beat
- 2. **V**: Ventricular ectopic (PVC)
- 3. **S**: Supraventricular ectopic (PAC)
- 4. **Q**: All beats

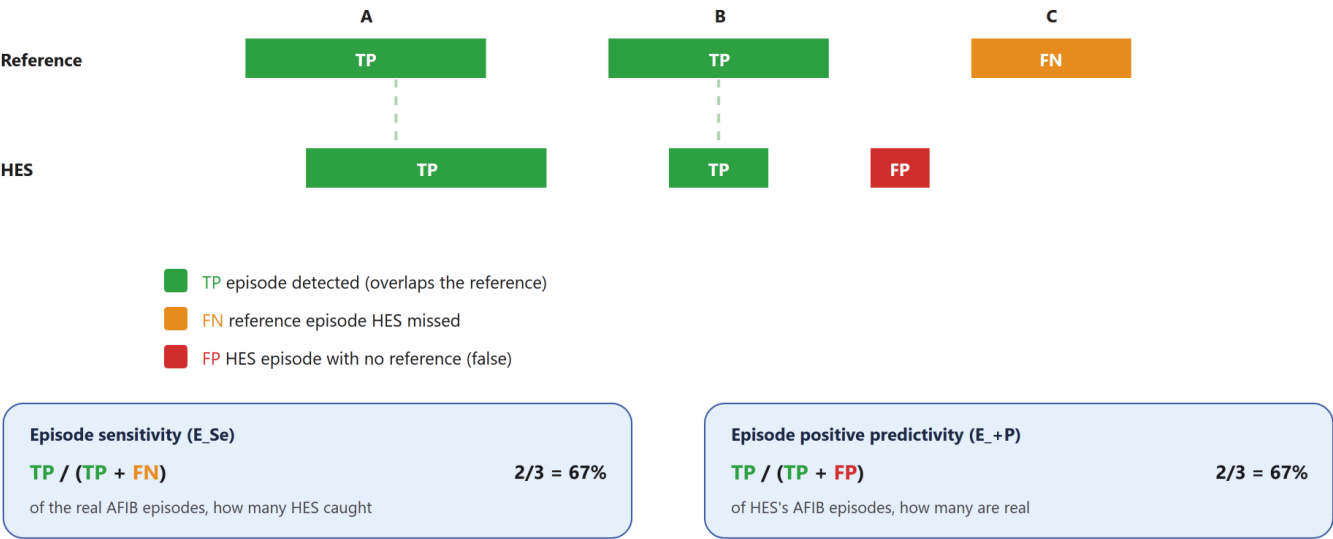
Metric	Formula	Meaning
Q_Se	$TP / (TP + FN)$	Sensitivity for all beats
Q_+P	$TP / (TP + FP)$	Positive Predictivity for all beats
V_Se	$VTP / (VTP + VFN)$	Sensitivity for V beats
V_+P	$VTP / (VTP + VFP)$	Positive Predictivity for V beats
S_Se	$STP / (STP + SFN)$	Sensitivity for S beats
S_+P	$STP / (STP + SFP)$	Positive Predictivity for S beats

AFIB Event Detection

AFIB episodes are matched against technician annotations using TP/FP/FN.

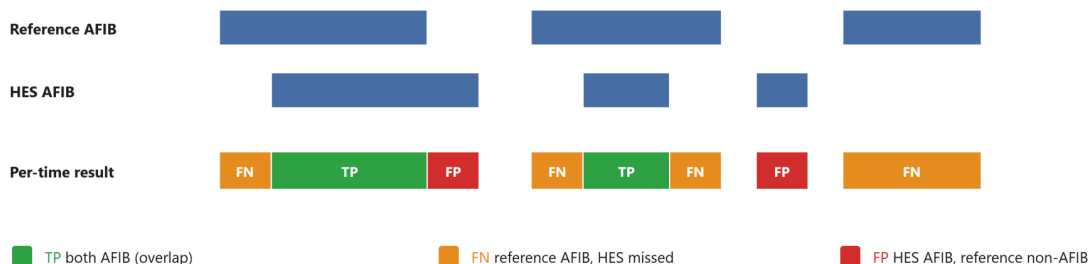
An episode is a TP if it overlaps a reference episode, FN if missed, and FP if false. Metrics: E_Se and E_+P.

AFIB Event Detection - Episode



Duration — every second is TP / FN / FP. → **D_Se, D_+P.**

AFIB Event Detection - Duration



Duration sensitivity (D_Se)

$$TP / (TP + FN)$$

$$28/62 = 45\%$$

of the real AFIB time, how much HES covered

Duration positive predictivity (D_+P)

$$TP / (TP + FP)$$

$$28/40 = 70\%$$

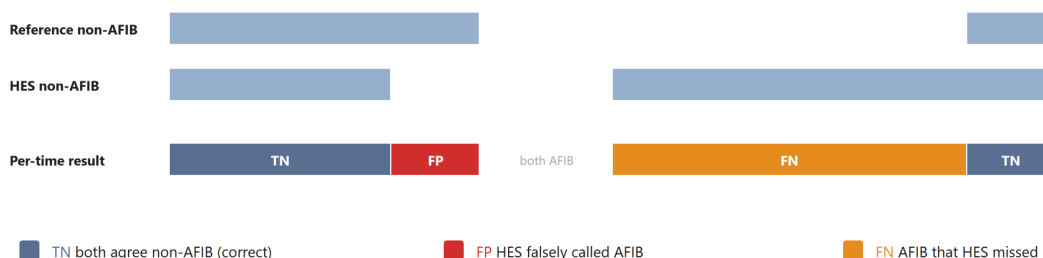
of HES's AFIB time, how much is real

Metric	Level	Formula	Meaning
E_Se	Episode	Detected / Reference	AFIB episodes detected
E_+P	Episode	True detected / Detected	Detected episodes that are true
D_Se	Duration	$TP / (TP + FN)$	True AFIB time detected
D_+P	Duration	$TP / (TP + FP)$	Detected AFIB time that is true

Specificity & NPV — the non-AFIB side

The run also evaluates non-AFIB time (TN). Specificity uses reference non-AFIB time (FP + TN), while NPV uses HES non-AFIB time (FN + TN).

AFIB Specificity and NPV



Specificity (D_Spec)

$$TN / (TN + FP)$$

$$35/45 = 78\%$$

of the reference non-AFIB time (TN + FP)

NPV (D_NPV)

$$TN / (TN + FN)$$

$$35/75 = 47\%$$

of the HES non-AFIB time (TN + FN)

Metric	Formula	Meaning
D_Spec — duration specificity	$TN / (TN + FP)$	of the true non-AFIB time, how much HES correctly left non-AFIB
D_NPV — duration negative predictive value	$TN / (TN + FN)$	of HES's non-AFIB time, how much is truly non-AFIB

Gross (pooled), not averaged

Every number in the report is **gross (pooled)**: sum the raw counts across **all** records first, then take **one** ratio — never the average of per-record percentages. Averaging would let a 20-beat record count the same as a 2,000-beat one, and a record with no reference beats has no percentage to average at all.

Example — **Q_Se = TP / (TP + FN)** over two records:

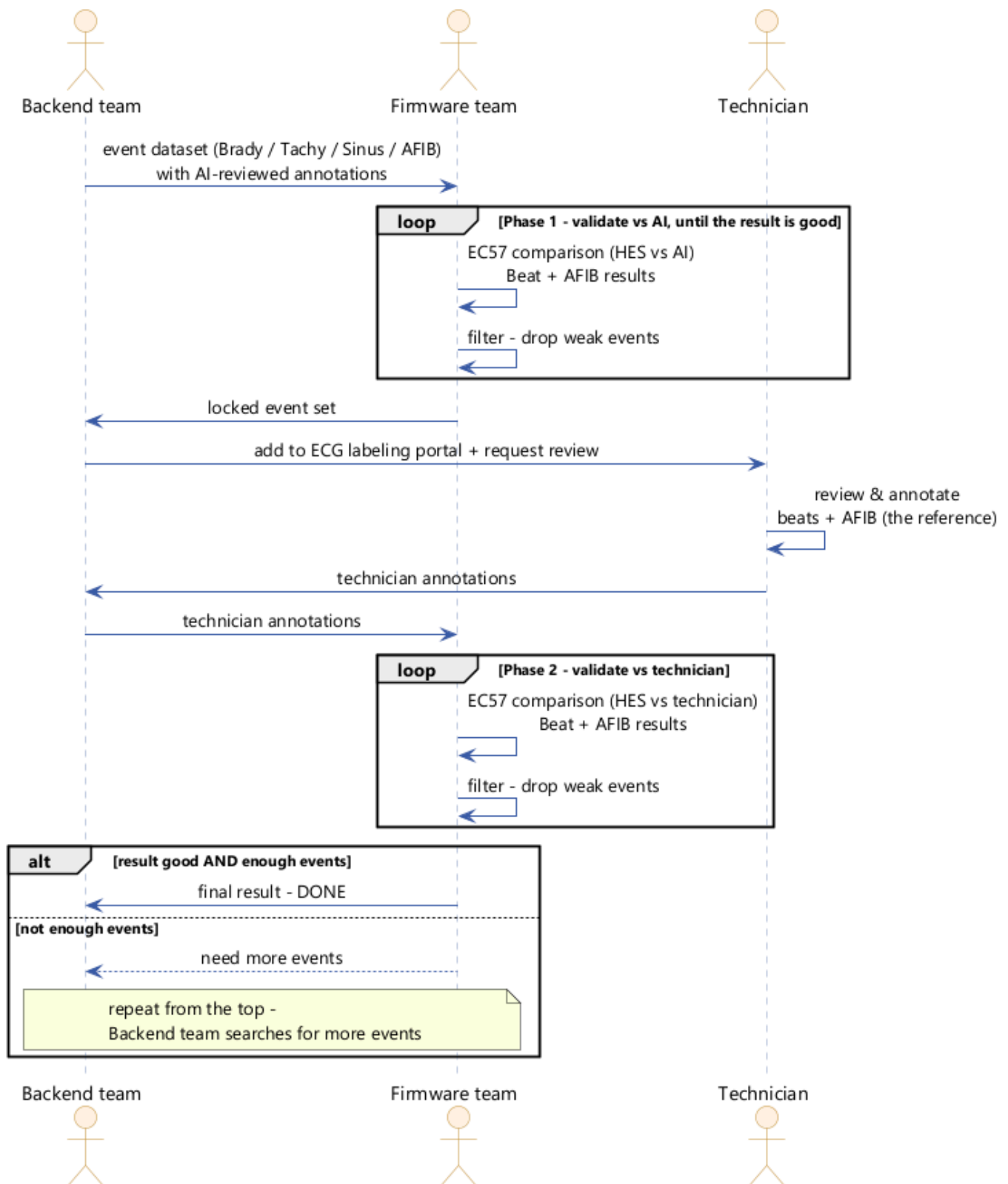
Record	TP	FN	per-record Q_Se
A	90	10	90.0%
B	990	10	99.0%
Gross	1,080	20	1,080 / 1,100 = 98.2%

The **average** of the two percentages is 94.5%, but the **gross** is **98.2%** — record B's 1,000 beats carry more weight. Every metric pools this way: beats sum TP / FN / FP; AFIB duration and Spec/NPV sum the AFIB seconds and TN / FP / FN. This is exactly why the **metric merge** and **filter / greedy** tabs re-pool the raw counts instead of averaging percentages.

High level workflow

Validating the HES algorithm for FDA submission is an **iterative** process between the **backend team**, the **firmware team**, and a **technician**.

HES validation - high-level process



Dataset

Full dataset (current)

Hosted on SharePoint: [full-dataset](#).

12,870 events in total — 1,452 already technician-labeled and 11,418 not yet technician-labeled. Grouped by reference source (Tech / AI), rhythm, and FBD:

1	Group	Rhythm	FBD	Event count	SourceVerify	FilterByDetectedSample	CalibTime	BeatErr
2	Tech	AFIB	false	222	Tech	FALSE	0	0.15
3	Tech	AFIB	true	479	Tech	TRUE	0	0.15
4	Tech	BRADY	false	159	Tech	FALSE	0	0.15
5	Tech	PAUSE	false	111	Tech	FALSE	0	0.15
6	Tech	SINUS	false	383	Tech	FALSE	0	0.15
7	Tech	TACHY	false	98	Tech	FALSE	0	0.15
8	AI	AFIB	false	11129	AI	FALSE	0	0.15
9	AI	BRADY	false	70	AI	FALSE	0	0.15
10	AI	PAUSE	false	90	AI	FALSE	0	0.15
11	AI	SINUS	false	8	AI	FALSE	0	0.15
12	AI	TACHY	false	121	AI	FALSE	0	0.15
13	TOTAL			12870				
14								

Selected 900 events for FDA

Hosted on SharePoint: [900-event set](#).

The validation set submitted for FDA — 900 events across five rhythm types:

	A	B	C
1	900-event validation set		
2	Type	Event count	
3	AFIB	300	
4	Brady	125	
5	Tachy	125	
6	Pause	50	
7	Sinus	300	
8	TOTAL	900	

Validate tab - EC57 comparison

cut ann ann to device event types validate metric merge filter hr report

Input data folder (eventid/ with ann/.dat/.hea/.json) W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\full-dataset\Tech\AFIB\FBD_false Browse

Output folder W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\validate Browse

Output file name (auto from config) validate_Tech_FBD-FALSE_calib0_beat0.15_rmlblank-T.xlsx

SourceVerify (reference) Tech

FilterByDetectedSample ☐

CalibTime (% used when FBD=false) 0

BeatErr (bub -w, seconds) 0.15

RemoveBlankAnn ☒

Add Spec/NPV to output ☒

Spec/NPV on AFIB records only ☒

Run Open output Open log

2026-08-12 16:16:53 AFIB_episode_Se 99

2026-08-12 16:16:53 AFIB_episode_P 97

2026-08-12 16:16:53 Blank .ann census 222

2026-08-12 16:16:53 Blank .ann files 0

2026-08-12 16:16:53 Non-blank .ann 222

2026-08-12 16:16:53 Blank ratio (%) 0.0000

2026-08-12 16:16:53 Cleanup intermediate artifacts 222

2026-08-12 16:16:53 cleanup: removed 222 per-hash folder(s) + excluded_records.csv

2026-08-12 16:16:53 DONE - output: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\validation-tool\output\metrics_summary.xlsx

2026-08-12 16:16:53 summary copied -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\validate\validate_Tech_FBD-FALSE_calib0_beat0.15_rmlblank-T.xlsx

2026-08-12 16:16:53 Spec/NPV added: AFIB_dur_Spec=81.48 AFIB_dur_NPV=84.83 -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\validate\validate_Tech_FBD-FALSE_calib0_beat0.15_rmlblank-T.xlsx

2026-08-12 16:16:53 report view added (sheets 'report' + 'mail_table') -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\validate\validate_Tech_FBD-FALSE_calib0_beat0.15_rmlblank-T.xlsx

2026-08-12 16:16:53 DONE

2026-08-12 16:16:53 Log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-12_validate.log (use 'Open log')

Report table (select cells + Ctrl+C or the buttons):

Strip Count	Q_Se	Q_P	V_Se	V_P	S_Se	S_P	E_Se	E_P	D_Se	D_P	D_Spec	D_NPV
222	96.43%	99.40%	78.97%	14.43%	42.30%	6.47%	99.00%	97.00%	95.00%	93.00%	81.48%	84.83%

Copy table Save .csv

1 Input
event folders (.dat/.hea/.json/.ann)

2 Output
destination folder + .xlsx file name

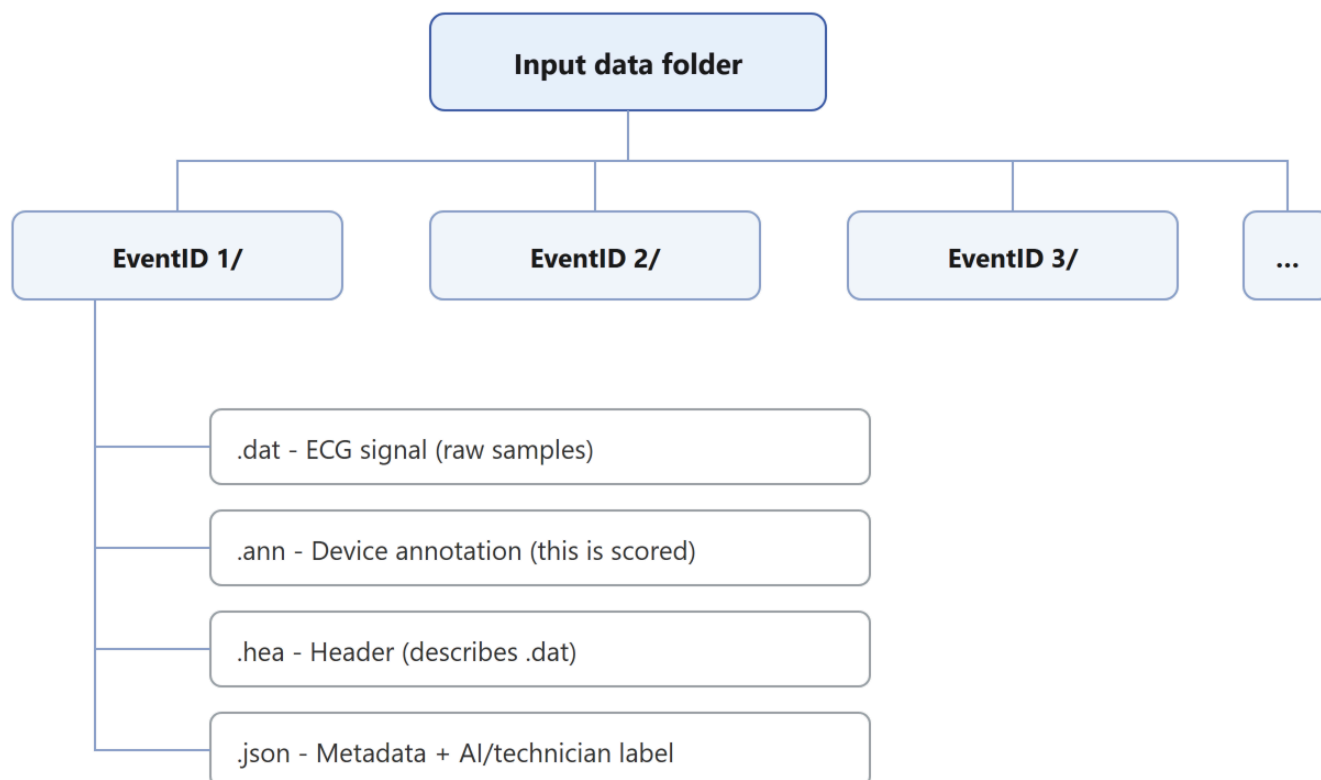
3 Parameters
reference (Tech / AI), FBD,
CalibTime, BeatErr, Spec/NPV

4 Result
the EC57 metrics
(report table under the log)

Fill the fields, then press **Run**. The metrics land in the report table under the log and are written to the output **metrics_summary.xlsx**.

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Input



Each **<eventId>/** sub-folder must contain the **four** files below:

File	What it holds
.dat	Record file — the raw ECG samples (200 Hz) for the event
.hea	Header file — record name, sample rate, format, sample count; describes the .dat
.json	Metadata shown on the ECG labeling portal — the AI / technician beats + AFIB annotation
.ann	The beats + AFIB annotation produced by the HES algorithm running on the device

Parameters

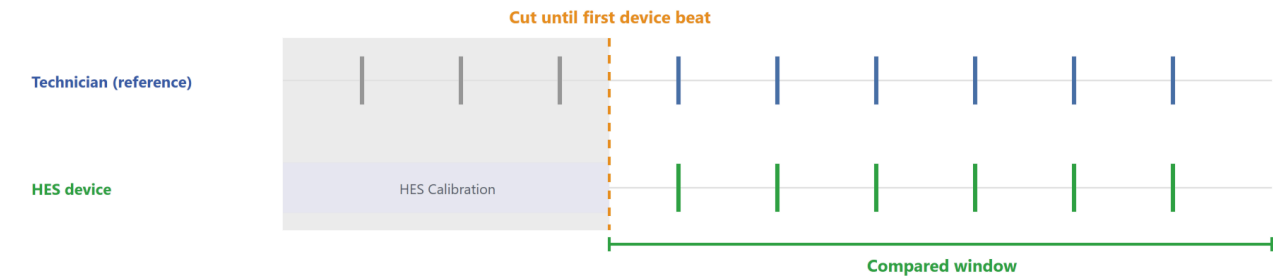
Defaults in brackets. The three timing-related fields are illustrated below.

Parameter	Default	Description
SourceVerify	AI	Reference: Tech or AI
FilterByDetectedSample (FBD)	on	Score only within the device detection window
CalibTime	0	Skip first N seconds; used when FBD is off
BeatErr	0.15	Beat-match window for bx , in seconds
RemoveBlankAnn	on	Exclude events with no detection
Add Spec/NPV to output	on	Add D_Spec / D_NPV columns
Spec/NPV on AFIB records only	on	Calculate Spec/NPV using AFIB records only

FilterByDetectedSample (FBD)

When the original .ann cannot be found, the event is re-run on the device, which detects nothing during its **calibration warm-up**. **FBD = on** trims the reference up to the device's **first detected beat**, so the warm-up is not scored as missed beats (**CalibTime** unused); **FBD = off** scores the full strip. Details: [ecgSD Firmware Workflow and Usage Guide](#).

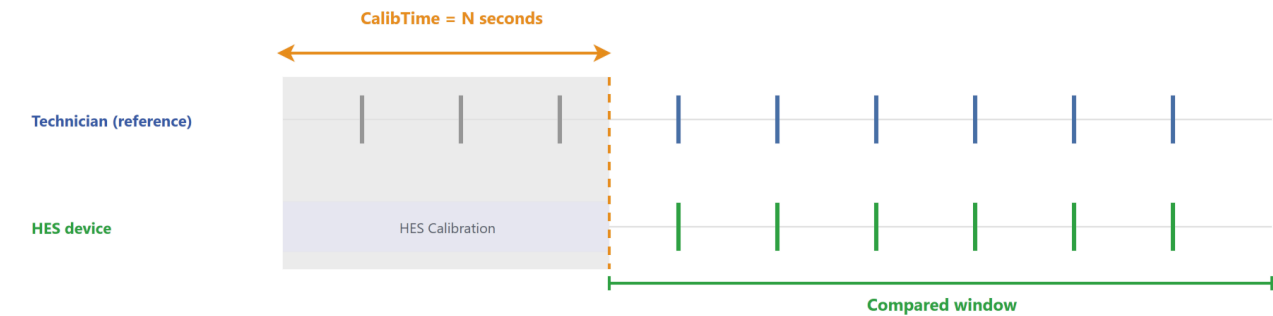
FilterByDetectedSample (FBD)



CalibTime

skip the **first N seconds** of both reference and device. 0 keeps all.

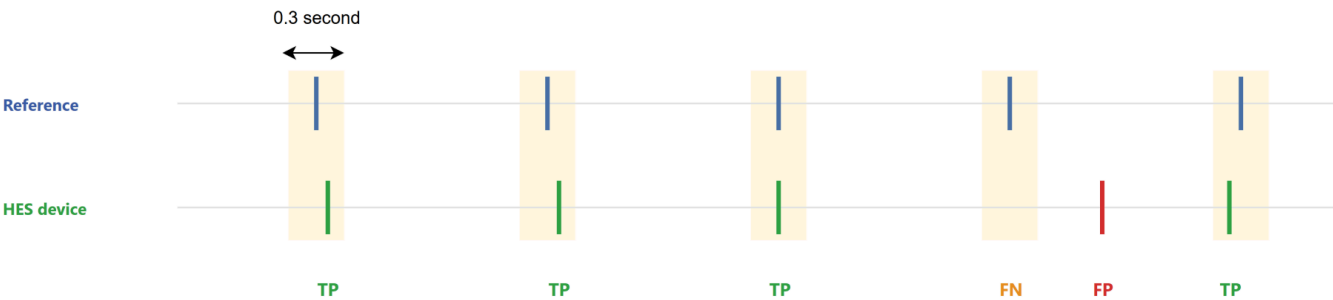
CalibTime (seconds)



BeatErr

The bxb tolerance matches a device beat to a reference beat within $\pm$ BeatErr seconds. Otherwise, the device beat is an FP and the reference beat is an FN. The default BeatErr is **0.15 s**, based on WFDB 10.7.0.

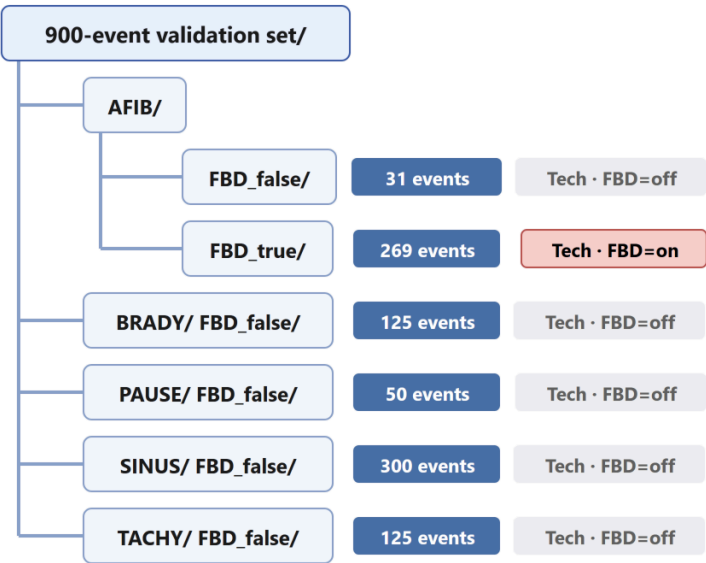
BeatErr (Default +- 0.15 second)



Example - Validate Brady dataset

The 900-event dataset is the merged result of six smaller datasets, as shown below:

900-event validation set



This example validates the Brady dataset from the selected 900-event dataset. Follow the steps below:

0. **Download** the **Brady** brady [900_event_source](#),

1 **Input**
BRADY/FBD_false

2 **Output folder**
test-data/metric-merge

3 **Auto-filled**
validate_config.json

4 **Run / Open output**
Run, then Open output

5 **Result / Copy table**
metrics; Copy table

cut ann ann to device event types validate metric merge filter hr report

Input data folder (eventid/ with .ann/.dat/.hea/.json) W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge\input-old-97\BRADY\FBD_false [Browse](#)

Output folder W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge [Browse](#)

Output file name (auto from config) brady.xlsx

SourceVerify (reference) Tech

FilterByDetectedSample ☐

CalibTime (s, used when FBD=false) 0

BeatErr (bxb -w, seconds) 0.15

RemoveBlankAnn ☒

Add Spec/NPV to output ☒

Spec/NPV on AFIB records only ☒

Run Open output Open log

2026-08-14 11:30:16 AFIB_dur_Se -
2026-08-14 11:30:16 AFIB_dur_+P -
2026-08-14 11:30:16 AFIB_episode_Se -
2026-08-14 11:30:16 AFIB_episode_+P -

2026-08-14 11:30:16 === Blank .ann census ===
2026-08-14 11:30:16 Total .ann files 125
2026-08-14 11:30:16 Blank .ann files 0
2026-08-14 11:30:16 Non-blank .ann 125
2026-08-14 11:30:16 Blank ratio (%) 0.0000

2026-08-14 11:30:16 === Cleanup intermediate artifacts ===
2026-08-14 11:30:16 cleanup: removed 125 per-hash folder(s) + excluded_records.csv

2026-08-14 11:30:16 =====
2026-08-14 11:30:16 DONE - output: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\validation-tool\output\metrics_summary.xlsx
2026-08-14 11:30:16 summary copied -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge\brady.xlsx
2026-08-14 11:30:16 Spec/NPV added: AFIB_dur_Spec=None AFIB_dur_NPV=None -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge\brady.xlsx
2026-08-14 11:30:16 report view added (sheets 'report' + 'mail_table') -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge\brady.xlsx

2026-08-14 11:30:17 === DONE ===
2026-08-14 11:30:17 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-14_validate.log (use 'Open log')

Report table (select cells + Ctrl+C, or the buttons):

Strip Count	Q_Se	Q_+P	V_Se	V_+P	S_Se	S_+P	E_Se	E_+P	D_Se	D_+P	D_Spec	D_NPV
125	94.68%	96.79%	82.77%	16.43%	61.60%	15.71%	N/A	N/A	N/A	N/A	N/A	N/A

Copy table Save.csv

- 1. **Input:** set the **Input data folder** to the Brady folder.
- 2. **Output folder:** a real project path such as `..\test-data\metric-merge` (not a temp folder); the file name auto-fills.
- 3. **Auto-filled:** the folder's `validate_config.json` fills **SourceVerify = Tech · FBD = off · CalibTime = 0 · BeatErr = 0.15 · RemoveBlankAnn = on**.
- 4. **Run / Open output:** press **Run**; when it finishes, **Open output** opens the saved **brady.xlsx**.
- 5. **Result / Copy table:** the EC57 metrics appear in the report table; **Copy table** copies them.

Open **brady.xlsx** (via **Open output**): it has **6 sheets** (the **summary** sheet is shown):

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	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Q_Se	Q_+P	S_Se	S_+P	V_Se	V_+P	AFIB_dur_Se	AFIB_dur_+P	AFIB_episode_Se	AFIB_episode_+P	AFIB_dur	AFIB_dur_NPV	
2		94.68	96.79	61.6	15.71	82.77	16.43	-	-	-	-	-	
3													
4	Total .ann files (input/<hash>/)	125											
5	Blank .ann files (0 beats)	0											
6	Non-blank .ann files	125											
7	Blank .ann ratio (%)	0											
8													
9	Manual exclusion breakdown	count	% of input/										
10	Total records in input/	125											
11	Excluded — Low Amplitude	0	0										
12	Excluded — Wrong Label	0	0										
13	Excluded — Miss AFib	0	0										
14	Excluded — Other Symptoms	0	0										
15	Excluded — Low AFib Duration	0	0										
16													
17													
18													
19													
20													

- **summary** — the gross metrics (one row).
- **report / mail_table** — the report-table view shown under the log.
- **bxline_sumstats** — per-record beat counts (Q / V / S) behind Q_Se, V_Se, S_Se.
- **epicmp_sumstats** — per-record AFIB episode + duration counts behind E_Se / D_Se and AFIB_dur_Spec / NPV.
- **excluded_records** — records dropped from scoring (if any).

The two per-record sheets are exactly what **metric merge** re-pools across groups.

Merge tab

As noted above, the selected **900-event** set is the **merge of six sub-sets** (one per rhythm × FBD). This tab pools them into one gross result.

Download all six sub-sets from SharePoint and **validate** each one → six metrics **.xlsx** (each like the Brady run above). Then follow the markers 1-4:

1 Inputs
6 group .xlsx

2 Run

3 Pooled
6 groups -> 900

4 Result
900 - 97.04

cut ann

ann to device

event types

validate

metric merge

filter

hr report

Inputs (metrics .xlsx files, or a folder of them)

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\brady.xlsx

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\afib_fbd_false.xlsx

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\afib_fbd_true.xlsx

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\pause.xlsx

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\sinus.xlsx

Add files... Add folder... Remove selected Clear

Output .xlsx

C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\merged_900.xlsx

Browse

Add Spec/NPV to output

☒

Spec/NPV on AFIB records only

☒

Run

Open output

Open log

2026-08-13 17:39:14 =====

2026-08-13 17:39:14 == RUN @ 2026-08-13 17:39:14 ==

2026-08-13 17:39:14 [brady] bxb=125 epicmp=125

2026-08-13 17:39:14 [afib_fbd_false] bxb=31 epicmp=31

2026-08-13 17:39:14 [afib_fbd_true] bxb=269 epicmp=269

2026-08-13 17:39:14 [pause] bxb=50 epicmp=50

2026-08-13 17:39:14 [sinus] bxb=300 epicmp=300

2026-08-13 17:39:14 [tachy] bxb=125 epicmp=125

2026-08-13 17:39:14 == MERGED (900 record) ==

2026-08-13 17:39:14 Q_Se 97.04 Q_+P 98.43 | V_Se 82.41 V_+P 14.74 | S_Se 23.39 S_+P 9.85

2026-08-13 17:39:14 E_Se 96.5 E_+P 97.31 | D_Se 95.54 D_+P 95.59 | D_Spec 85.68 D_NPV 85.54

2026-08-13 17:39:14 Saved -> C:\Users\haupham\AppData\Local\Temp\claude\w--report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\merged_900.xlsx

2026-08-13 17:39:14 == DONE ==

2026-08-13 17:39:14 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-13_metric_merge.log (use 'Open log')

Report table (select cells + Ctrl+C, or the buttons):

Strip Count	Q_Se	Q_+P	V_Se	V_+P	S_Se	S_+P	E_Se	E_+P	D_Se	D_+P	D_Spec	D_NPV
900	97.04%	98.43%	82.41%	14.74%	23.39%	9.85%	96.50%	97.31%	95.54%	95.59%	85.68%	85.54%

Copy table Save.csv

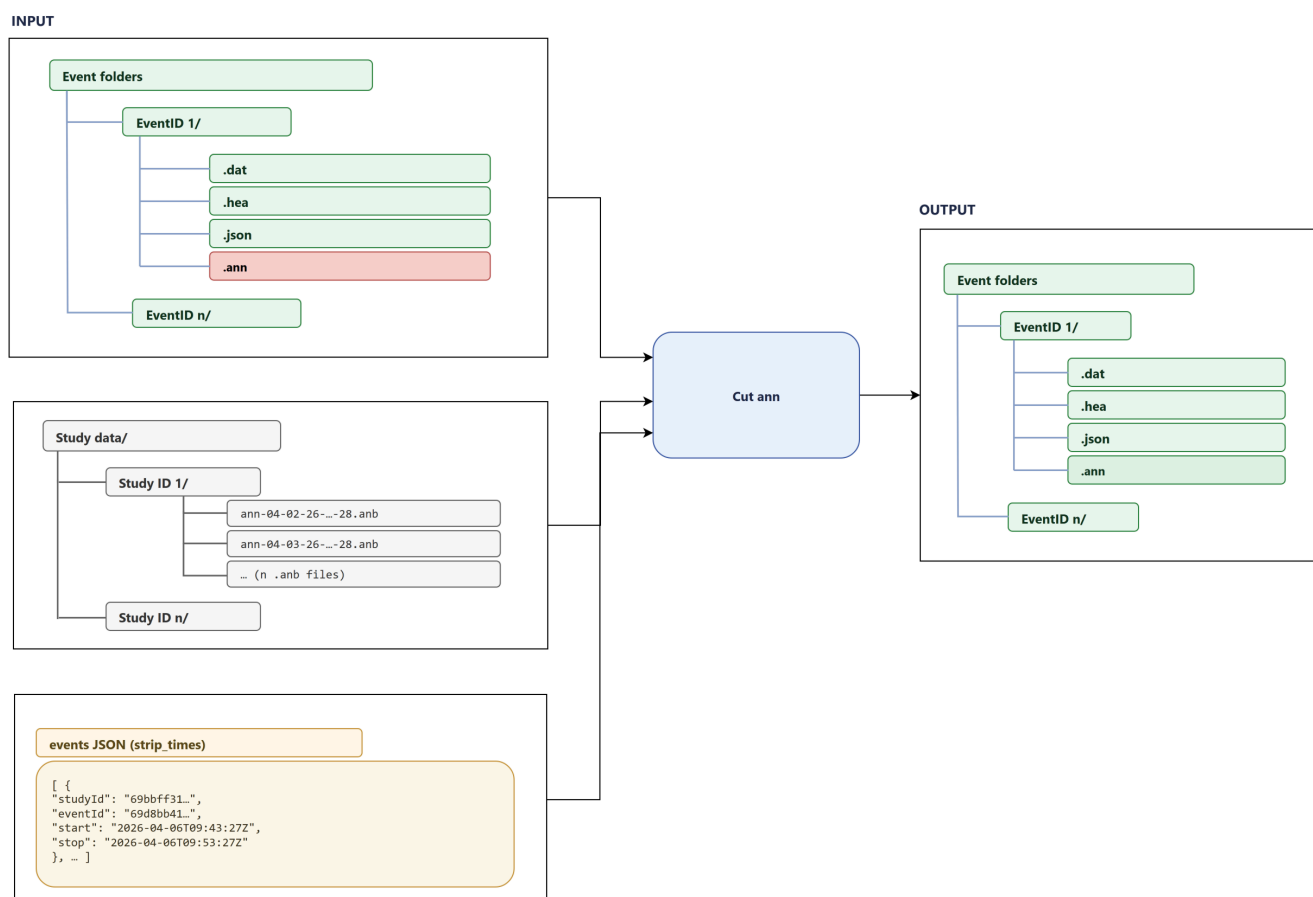
1. **Inputs** — Add the six **.xlsx** files or select a folder containing them.
2. **Run** — Merge the datasets.
3. **Pooled** — Verify the merged counts total 900 events.
4. **Result** — View the pooled EC57 metrics; **Copy table** to copy the results.

Pooling the six groups gives the locked EC57 result for the 900 set:

Strip	Q_Se	Q_+P	V_Se	V_+P	S_Se	S_+P	E_Se	E_+P	D_Se	D_+P	D_Spec	D_NPV
900	97.04	98.43	82.41	14.74	23.39	9.85	96.5	97.31	95.54	95.59	85.68	85.54

Cut ann tab

When BE provides the event folders, the **.ann** files are missing because annotations are stored in the study's **.anb** file. **Cut ann** rebuilds each event folder using the **.anb** file and event JSON (**studyId**, **start**, **stop**).



Run it:

0. **Download** the cut-ann sample from [SharePoint](#) and extract it to **..\test-data\cut-ann**.

- 1 JSON folder
..\\afib\\json
- 2 Study data folder
..\\afib\\anb
- 3 Output folder
the Event folders
- 4 Run
slices .ann + checks

cut ann	ann to device	event types	validate	metric merge	filter	hr report
JSON folder (events)	W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\cut-ann\afib\json					Browse
Study data folder (.anb)	W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\cut-ann\afib\anb					Browse
Output folder	W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\cut-ann\afib\event					Browse

Run	Open output	Open log	Remove incomplete
2026-08-14 14:21:23 [ok]	6a3f7ed82694cc0001d1fe5b: ann-06-12-26-12-12-05-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-14-26-02-03-00-00.ann (threshold=2211000, kept=791, dropped=215214)		
2026-08-14 14:21:23 [ok]	6a3a92a756915d000166e8e1: ann-06-13-26-06-21-59-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-13-26-17-59-09-00.ann (threshold=5486000, kept=1305, dropped=74090)		
2026-08-14 14:21:23 [ok]	6a4c1c2bf7f159000114c05a: ann-06-19-26-15-04-30-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-20-26-04-06-58-00.ann (threshold=6509600, kept=1367, dropped=50441)		
2026-08-14 14:21:23 [ok]	6a4237161f60b0001b0377e: ann-06-23-26-11-54-21-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-23-26-21-42-44-00.ann (threshold=3466500, kept=836, dropped=92365)		
2026-08-14 14:21:23 [ok]	6a3b375eaa04c90001cc0b1f: ann-06-21-26-08-32-12-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-21-26-17-22-02-00.ann (threshold=2758000, kept=817, dropped=148840)		
2026-08-14 14:21:23 [ok]	6a3f7ed82694cc0001d1fe5b: ann-06-14-26-12-03-04-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-14-26-23-08-11-00.ann (threshold=2941400, kept=912, dropped=178447)		
2026-08-14 14:21:23 [ok]	6a43b0fccc7b04000193608f: ann-05-29-26-12-03-32-20.ann -> 0m-0m_x2_0_Calib1_data-all-05-30-26-01-56-24-00.ann (threshold=6394400, kept=936, dropped=243375)		
2026-08-14 14:21:23 [ok]	6a32f65f307d40001c041b: ann-06-12-26-23-09-20-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-13-26-04-38-13-00.ann (threshold=1866500, kept=981, dropped=283208)		
2026-08-14 14:21:23 [ok]	6a341a6a366ff800018e9b05: ann-06-17-26-23-32-56-12.ann -> 0m-0m_x2_0_Calib1_data-all-06-18-26-05-43-00-00.ann (threshold=2280800, kept=786, dropped=51028)		
2026-08-14 14:21:23 [ok]	6a33af3400a91d0001d53ef2: ann-06-16-26-13-24-48-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-17-26-15-39-40-00.ann (threshold=13858400, kept=1144, dropped=101242)		
2026-08-14 14:21:24 [ok]	6a0c2cf5366f61000115f5b4: ann-05-03-26-14-23-58-28.ann -> 0m-0m_x2_0_Calib1_data-all-05-04-26-04-39-14-00.ann (threshold=5223200, kept=1078, dropped=76599)		
2026-08-14 14:21:24 [ok]	6a0c2cf5366f61000115f5b4: ann-05-07-26-19-39-43-28.ann -> 0m-0m_x2_0_Calib1_data-all-05-08-26-11-46-15-00.ann (threshold=6558400, kept=1478, dropped=36934)		
2026-08-14 14:21:24 [ok]	6a3b375eaa04c90001cc0b1f: ann-06-21-26-08-32-12-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-22-26-08-06-42-00.ann (threshold=13374000, kept=941, dropped=149816)		
2026-08-14 14:21:24 [ok]	6a341f69366ff800018e9b05: ann-06-17-26-23-32-56-12.ann -> 0m-0m_x2_0_Calib1_data-all-06-18-26-10-19-22-00.ann (threshold=5597200, kept=646, dropped=51888)		
2026-08-14 14:21:24 [ok]	6a0c2cf5366f61000115f5b4: ann-05-03-26-14-23-58-28.ann -> 0m-0m_x2_0_Calib1_data-all-05-04-26-04-39-14-00.ann (threshold=5223200, kept=1078, dropped=76599)		
2026-08-14 14:21:24 [ok]	6a3a92a756915d000166e8e1: ann-06-13-26-06-21-59-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-22-26-01-59-50-00.ann (threshold=8996000, kept=1728, dropped=135933)		
2026-08-14 14:21:24 [ok]	6a3d7ea9077813000100769f: ann-06-21-26-09-30-37-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-22-26-02-55-29-00.ann (threshold=9658400, kept=1815, dropped=135838)		
2026-08-14 14:21:24 [ok]	6a464926c77b040001937798: ann-06-21-26-08-05-33-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-22-26-12-39-18-00.ann (threshold=1845800, kept=1330, dropped=497358)		
2026-08-14 14:21:24 [ok]	6a48096c20c3c200013e1d54: ann-06-19-26-15-10-56-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-19-26-20-17-40-00.ann (threshold=800800, kept=1244, dropped=38766)		
2026-08-14 14:21:25 [ok]	6a4213a6d7914300018d1c5c: ann-06-23-26-12-48-11-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-25-26-11-56-48-00.ann (threshold=31063400, kept=1258, dropped=365407)		
2026-08-14 14:21:25 [ok]	6a42cc0bebe7f00010c609b: ann-06-25-26-21-27-15-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-26-26-12-49-00-00.ann (threshold=8181000, kept=1248, dropped=107597)		
2026-08-14 14:21:25 [ok]	6a42a02ee264f70001cad14c: ann-06-26-26-23-14-58-12.ann -> 0m-0m_x2_0_Calib1_data-all-06-27-26-13-21-35-00.ann (threshold=7999400, kept=973, dropped=68982)		
2026-08-14 14:21:25 [ok]	6a4478710e08210001d0b469a: ann-06-23-26-16-14-04-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-24-26-08-19-32-00.ann (threshold=785600, kept=832, dropped=63761)		
2026-08-14 14:21:25 [ok]	6a44b0487e419e00015c7023: ann-06-29-26-10-57-40-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-30-26-15-26-15-00.ann (threshold=15463000, kept=747, dropped=96121)		
2026-08-14 14:21:25 [ok]	6a96f780447f938001d4e547: ann-06-29-26-13-30-18-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-30-26-03-28-52-00.ann (threshold=2380280, kept=907, dropped=143884)		
2026-08-14 14:21:25 [ok]	6a75934a00c60001d404bc: ann-06-30-26-23-13-03-16.ann -> 0m-0m_x2_0_Calib1_data-all-07-01-26-04-26-49-00.ann (threshold=885200, kept=856, dropped=48958)		
2026-08-14 14:21:25 ... 150/158 studies			
2026-08-14 14:21:25 [ok]	6a4602db05e920001e96635: ann-06-24-26-15-38-39-16.ann -> 0m-0m_x2_0_Calib1_data-all-06-25-26-02-46-31-00.ann (threshold=5134400, kept=1025, dropped=106499)		
2026-08-14 14:21:25 [ok]	6a47424c2660e00014587d1: ann-06-29-26-15-32-28-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-30-26-05-44-29-00.ann (threshold=6624200, kept=968, dropped=111657)		
2026-08-14 14:21:25 [ok]	6a474e905002a00017b2d8e: ann-06-24-26-16-08-37-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-25-26-01-22-59-00.ann (threshold=3052400, kept=958, dropped=178308)		
2026-08-14 14:21:25 [ok]	6a47089502cec30001463b2f: ann-06-25-26-15-21-16-28.ann -> 0m-0m_x2_0_Calib1_data-all-06-26-26-12-15-17-00.ann (threshold=10008200, kept=887, dropped=117801)		
2026-08-14 14:21:25 [ok]	6a4f28c122bf100001de5fd: ann-06-25-26-16-16-13-20.ann -> 0m-0m_x2_0_Calib1_data-all-06-26-26-11-07-45-00.ann (threshold=9978400, kept=838, dropped=94580)		
2026-08-14 14:21:25 [ok]	6a4e4aaa15579000195f21d: ann-07-06-26-08-16-51-28.ann -> 0m-0m_x2_0_Calib1_data-all-07-07-26-01-13-20-00.ann (threshold=7157800, kept=807, dropped=65712)		
2026-08-14 14:21:25 [ok]	6a4e072de55e3a0001e08f88: ann-07-01-26-23-22-39-12.ann -> 0m-0m_x2_0_Calib1_data-all-07-02-26-13-21-49-00.ann (threshold=7910000, kept=474, dropped=42329)		
2026-08-14 14:21:25 ... 150/158 studies			
2026-08-14 14:21:25 Done, events ok: 178, failed: 5			
2026-08-14 14:21:26 WARNING: 5/183 folder(s) do NOT have all 4 files (.ann/.dat/.hea/.json):			
2026-08-14 14:21:26 WARN 69d8131795f06100019e40c9: missing .ann			
2026-08-14 14:21:26 WARN 69d05a2464c5300019f5a5d: missing .ann			
2026-08-14 14:21:26 WARN 69e638177ed09e00010c0887: missing .ann			
2026-08-14 14:21:26 WARN 6a03f85e74c090001be4085: missing .ann			
2026-08-14 14:21:26 WARN 6a1028f32f03000129c500: missing .ann			
2026-08-14 14:21:26 Errors logged to W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\cut-ann\afib\event\error.txt (5 entries)			
2026-08-14 14:21:26 == DONE ==			
2026-08-14 14:21:26 Log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-tool\logs\2026-08-14_cut_ann_log (use 'Open log')			

1. **JSON folder** = ..\test-data\cut-ann\afib\json
2. **Study data folder** = ..\test-data\cut-ann\afib\anb
3. **Output folder** = Event folders in ..\test-data\cut-ann\afib\event containing .dat/.hea/.json; Cut ann adds the .ann file.
4. **Run** — Cuts the .ann for each event, renames all four files to the same <base>, and updates the .hea record name. **Remove incomplete** deletes folders missing any of the 4 files.

Ann to device tab

ann to device copies the device's beats and AFIB from **.ann** to the event JSON's **aiPrediction**, replacing the AI result. This allows the technician to review the device's actual detection on the ECG portal.

ann to device — replace the AI result with the device result



It writes beats, **hesBeatStatus**, and AFIB **rhythm** runs into **aiPrediction**.

Run it:

0. **Download** the sample from [SharePoint](#) and extract it to ..\test-data\ann-to-device\input.

- 1 Source folder
..ann-to-device\input
- 2 Output folder
..ann-to-device\output
- 3 Run
write .ann → aiPrediction
- 4 Result
Converted 5/5 events

cut ann ann to device event types validate metric merge filter hr report

Source folder W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\ann-to-device\input Browse

Output folder W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\ann-to-device\output Browse

Run Open output Open log

2026-08-14 14:32:32 =====
2026-08-14 14:32:32 *** RUN @ 2026-08-14 14:32:32 ***
2026-08-14 14:32:32 Found 5 events in W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\ann-to-device\input
2026-08-14 14:32:32 ... 5/5
2026-08-14 14:32:32 Converted 5/5 events -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\ann-to-device\output
2026-08-14 14:32:32 total beats=3221 | total AFib runs (rhythm)=4
2026-08-14 14:32:32 *** DONE ***
2026-08-14 14:32:32 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-14_ann_to_device.log (use 'Open log')

1. Source folder = ..\test-data\ann-to-device\input (5 event folders with .ann/.dat/.hea/.json).
2. Output folder = any folder (e.g. ..\test-data\ann-to-device\output).
3. Run — writes device .ann data to each .json's aiPrediction.
4. Result — 5/5 events converted · 3,221 beats · 4 AFIB runs. Confirm — Re-run validate with SourceVerify = AI. All metrics should be 100% because the device .ann and aiPrediction contain the same device results.

Self-check
every metric = 100%

cut ann ann to device event types validate metric merge filter hr report

Input data folder (eventid/ with .ann/.dat/.hea/.json) W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\ann-to-device\input Browse

Output folder C:\Users\haupham\AppData\Local\Temp\claudew---report-HES-VALIDATION\7cde1046-aad-4bd9-9609-5160c943aca0\scratchpad\ex_out Browse

Output file name (auto from config) a2d_selfcheck.xlsx

SourceVerify (reference) AI

FilterByDetectedSample ☒

CalibTime (s, used when FBD=false) 0

BeatErr (bxb -w, seconds) 0.15

RemoveBlankAnn ☒

Add Spec/NPV to output ☒

Spec/NPV on AFIB records only ☒

Run Open output Open log

2026-08-14 14:32:38 AFIB_episode_Se 100
2026-08-14 14:32:38 AFIB_episode_P 100
2026-08-14 14:32:38 *** Blank .ann census ***
2026-08-14 14:32:38 Total .ann files 5
2026-08-14 14:32:38 Blank .ann files 0
2026-08-14 14:32:38 Non-blank .ann 5
2026-08-14 14:32:38 Blank ratio (%) 0.0000
2026-08-14 14:32:38 *** Cleanup intermediate artifacts ***
2026-08-14 14:32:38 Cleanup: removed 5 per-hash folder(s) + excluded_records.csv
2026-08-14 14:32:38 =====
2026-08-14 14:32:38 DONE - output: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\validation-tool\output\metrics_summary.xlsx
2026-08-14 14:32:38 summary copied -> C:\Users\haupham\AppData\Local\Temp\claudew---report-HES-VALIDATION\7cde1046-aad-4bd9-9609-5160c943aca0\scratchpad\ex_out\selfcheck.xlsx
2026-08-14 14:32:38 Spec/NPV added: AFIB_dur_Spec=100.0 AFIB_dur_NPV=100.0 -> C:\Users\haupham\AppData\Local\Temp\claudew---report-HES-VALIDATION\7cde1046-aad-4bd9-9609-5160c943aca0\scratchpad\ex_out\selfcheck.xlsx
2026-08-14 14:32:38 report view added (sheets 'report' + 'mail_table') -> C:\Users\haupham\AppData\Local\Temp\claudew---report-HES-VALIDATION\7cde1046-aad-4bd9-9609-5160c943aca0\scratchpad\ex_out\selfcheck.xlsx
2026-08-14 14:32:38 *** DONE ***
2026-08-14 14:32:38 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-14_validate.log (use 'Open log')

Report table (select cells + Ctrl+C, or the buttons):

Strip Count	Q_Se	Q_P	V_Se	V_P	S_Se	S_P	E_Se	E_P	D_Se	D_P	D_Spec	D_NPV
5	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Copy table Save .csv

Event types tab

Event types assigns each event one rhythm (AFIB, Brady, Tachy, Pause, or Sinus/Other) for dataset accounting.

Classification — Based on the **rhythm** and **interpretation** fields in the .json strip (technician label).

```
"strip": {
  "rhythm": [
    {
      "type": "AFIB",
      "isDeleted": false,
      "start": 47,
      "stop": 149942
    }
  ],
  "interpretation": "Atrial Fibrillation/Flutter"
}
```

AFIB: `strip.rhythm` contains a non-deleted **AFIB** run.

Pause / Tachy / Brady: detected by keywords in `strip.interpretation`.

If `strip.interpretation` is blank or "agree", use `aiPrediction.interpretation`.

The table below shows the rules with an example from the 900-event dataset.

Rhythm	Fires when	Example (real event)
AFIB	<code>strip.rhythm</code> has a run with type = AFIB, not deleted	rhythm run type = AFIB (from the field, not the text)
Pause	<code>strip.interpretation</code> contains pause	"Sinus Rhythm at 46 bpm and Pause 2.7s with SVE."
Tachy	<code>strip.interpretation</code> contains tachy	"Sinus Tachycardia, Ventricular Trigeminals, PVCs"
Brady	<code>strip.interpretation</code> contains brady	"Sinus Bradycardia with Sinus Arrhythmia and PACs"
Sinus/Other	none of the above fired	"3rd Deg AVB with SVEs, SVE Couplets, SVE Runs"

Multi-rhythm events: Apply priority **AFIB > Pause > Tachy > Brady** so each event belongs to one bucket.

Event interpretation (real)	Rules matched	Kept (priority)
"Bradycardia 29 bpm and Pause 4.4s; Atrial Fibrillation/Flutter 35-69; Sinus 52"	AFIB · Pause · Brady	AFIB
"2nd degree AV Block 26 bpm and Pause 2.5s; Bradycardia 27; Sinus 54"	Pause · Brady	Pause
" Atrial Fibrillation/Flutter 93-233; Sinus Tachycardia 144"	AFIB · Tachy	AFIB

Note: AFIB is determined by the **rhythm** field, not interpretation text. Only **Pause / Tachy / Brady** use text matching.

Run on the 900-event set using steps 1–4 below.

1 Input

the event folder

2 Output xlsx

the counts workbook

3 Run

4 Result

one rhythm per event

cut annann to deviceevent typesvalidatemetric mergefilterhr report

Event folder

W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\metric-merge\input-old-97

Browse

Output xlsx

C:\Users\haupham\AppData\Local\Temp\claudew\report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\event_types_900.xlsx

Browse

Single-label only (default outputs both, single-label first)

☐

Run

Open output

Open log

2026-08-14 17:19:23 =====

2026-08-14 17:19:23 === RUN @ 2026-08-14 17:19:23 ===

2026-08-14 17:19:23 [single-label] TOTAL events: 900 (one rhythm per event)

2026-08-14 17:19:23 AFIB 300

2026-08-14 17:19:23 Brady 125

2026-08-14 17:19:23 Tachy 125

2026-08-14 17:19:23 Pause 50

2026-08-14 17:19:23 Sinus 300

2026-08-14 17:19:23 (source) car5 130 / tech 770

2026-08-14 17:19:23 [multi-label] TOTAL events: 900

2026-08-14 17:19:23 Sinus/Other 300

2026-08-14 17:19:23 AFIB 249

2026-08-14 17:19:23 Brady 125

2026-08-14 17:19:23 Tachy 125

2026-08-14 17:19:23 Pause + Brady 37

2026-08-14 17:19:23 AFIB + Tachy 28

2026-08-14 17:19:23 Pause 12

2026-08-14 17:19:23 AFIB + Pause 7

2026-08-14 17:19:23 AFIB + Pause + Brady 6

2026-08-14 17:19:23 AFIB + Brady 5

2026-08-14 17:19:23 AFIB + Pause + Tachy 2

2026-08-14 17:19:23 AFIB + Tachy + Brady 2

2026-08-14 17:19:23 AFIB + Pause + Tachy + Brady 1

2026-08-14 17:19:23 Pause + Tachy 1

2026-08-14 17:19:23 [total] AFIB 300

2026-08-14 17:19:23 [total] Brady 176

2026-08-14 17:19:23 [total] Tachy 159

2026-08-14 17:19:23 [total] Pause 66

2026-08-14 17:19:23 [total] Sinus/Other 300

2026-08-14 17:19:23 Saved -> C:\Users\haupham\AppData\Local\Temp\claudew\report-HES-VALIDATION\7cde1046-aaad-4bd9-9609-5160c943aca0\scratchpad\ex_out\event_types_900.xlsx

2026-08-14 17:19:23 === DONE ===

2026-08-14 17:19:23 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-14_event_types.log (use 'Open log')

1. Input — Event folder = ..\test-data\metric-merge\input-old-97 (the folder holding the event groups).

2. Output xlsx = anywhere.

3. Run.

4. Result — TOTAL 900, one rhythm per event: AFIB 300 · Brady 125 · Tachy 125 · Pause 50 · Sinus/Other 300 = 900 (source car5 130 / tech 770).

The output .xlsx — a Summary with the counts plus one sheet per rhythm (friendlyId / recordHash / source / interpretation):

A1

Single-label (one rhythm per event)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Single-label														
2	Type	Event count													
3	AFIB	300													
4	Brady	125													
5	Tachy	125													
6	Pause	50													
7	Sinus	300													
8	TOTAL	900													
9	Source														
10	car5	130													
11	tech	770													
12															
13															
14															
15															
16															
17															
18															
19															
20															
21															

Summary

AFIB

Brady

Tachy

Pause

Sinus-Other

+

Filter tab

One tab with two curation modes:

- **greedy** — keeps the best **N** records using maximin.
- **flag** — removes records below a fixed metric threshold.

Both modes recompute metrics for the remaining records and output the **selected** set. **flag** also outputs a **flagged** sheet. Input: one or more validation **.xlsx** files.

Demo data

0. **Download** the demo pools from [SharePoint](#):

- **demo_filter.xlsx** — 10 records for **greedy** and all **flag** modes.
- **demo_ceil.xlsx** — 6 records for **Ceil** mode only.

Select the relevant file in **Inputs** before running.

Greedy

Mode = greedy removes records one by one until **N** remain. Each removal improves the weakest **Objective** metric, keeping the selected set balanced.

Field	Description
N	Number of records to keep.
Objective	Metrics to optimize: E_Se=0, E_+P=1, D_Se=2, D_+P=3, D_Spec=4, D_NPV=5.
Ceil E_Se	Optional upper cap for E_Se to prevent it from increasing too much.
Prefilter	Removes records with any metric ≤ 0 . Keep it enabled.

The tool fields are shown below.

cut annann to deviceevent typesvalidatemetric mergefilterhr report

Inputs: metrics .xlsx files (or a folder of them)

W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\filter\afib_FBD_true.xlsx

1Add files...

Add folder...

Remove selected

Clear

Mode

2greedy

Keep the best N records (maximin) — maximises the weakest metric of the Objective.

Filter (flag modes)

flag-Q: Q_Se <92.79

flag-Q: Q_+P <98.87

flag-V: V_Se <87.58

flag-V: V_+P <19.09

flag-low-Dse: DSe <90.0

3Greedy select

N — records to keep300

Objective^{spec-npv}

Ceil E_Se (empty = off)95.5

Prefilter (drop metric <= 0)☒

Output metrics summary .xlsx

W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\filter_greedy_300.xlsx

Run

4In output

Open log

1Add files → demo_filter.xlsx

2Mode = greedy

3set N / Objective / Ceil / Prefilter

4Run

Report table (select cells + Ctrl-C, or the buttons):

Copy table

Save .csv

Output:

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- **summary** — Selected count + 6 gross metrics.
- **epicmp_sumstats** — One row per selected record.
- **selected** — Hashes of selected records.

Prefilter: Removes records with any metric ≤ 0 or missing **D_Se** before selection. Keep it **on**.

Demo: **10** → **7** clean records. Real 701-AFIB pool: **701** → **575**.

Prefilter

Record	E_Se	E_+P	D_Se	D_+P	D_Spec	D_NPV	prefilter
Good1	100	100	95	96	99.0	98.8	✓ keep
Good2	100	100	94	95	98.4	98.0	✓ keep
LowSpec1	93.3	92.9	96	35	68.1	99.0	✓ keep
LowSpec2	92.9	92.3	95	34	69.5	98.9	✓ keep
LowQ1	100	100	90	90	98.0	98.0	✓ keep
LowV1	100	100	92	93	98.6	98.4	✓ keep
LowDSe1	100	100	55	85	97.3	91.5	✓ keep
DevMiss1	0	-	0	-	100	86.7	✗ drop — E_Se = 0
FalseAFIB1	-	0	-	0	93.3	100	✗ drop — no D_Se
AllZero1	0	0	0	0	88.9	88.9	✗ drop — all 0

10 records

→ drop 3

→ 7 clean

How greedy works: Each round:

1. Review the 6 metrics of the kept records.
2. Find the weakest **Objective** metric.
3. Remove the record that gives the largest improvement to that metric.
4. Repeat until **N** records remain.

The **Objective** defines which metrics are considered. No weights are used.

all6 — Watches all six metrics. In the demo, the weakest metric changes from **D_+P** (rounds 1–2) to **D_Se** (round 3).

objective all6 (keep N = 4)

Round 1 - 7 records

Input

metric	Good1	Good2	LowSpec1	LowSpec2	LowQ1	LowV1	LowDSe1	GROSS
E_Se	100	100	93.3	92.9	100	100	100	95.3
E_+P	100	100	92.9	92.3	100	100	100	95.1
D_Se	95	94	96	95	90	92	55	88.4
D_+P	96	95	35	34	90	93	85	65.2
D_Spec	99	98.4	68.1	69.5	98	98.6	97.3	89.5
D_NPV	98.8	98	99	98.9	98	98.4	91.5	97.3

weakest metric = D_+P 65.2

Recompute D_+P if that record is dropped

drop →	Good1	Good2	LowSpec1	LowSpec2	LowQ1	LowV1	LowDSe1
D_+P becomes	61.3	60.3	74.7★	74.5	62.6	62.3	63.3

drop LowSpec1 — D_+P 65.2 → 74.7

Round 2 - 6 records

Input

metric	Good1	Good2	LowSpec2	LowQ1	LowV1	LowDSe1	GROSS
E_Se	100	100	92.9	100	100	100	96.4
E_+P	100	100	92.3	100	100	100	96.3
D_Se	95	94	95	90	92	55	87.4
D_+P	96	95	34	90	93	85	74.7
D_Spec	99	98.4	69.5	98	98.6	97.3	93.2
D_NPV	98.8	98	98.9	98	98.4	91.5	97.1

weakest metric = D_+P 74.7

Recompute D_+P if that record is dropped

drop →	Good1	Good2	LowSpec2	LowQ1	LowV1	LowDSe1
D_+P becomes	71	70	92.3★	72.5	72.1	73.4

drop LowSpec2 — D_+P 74.7 → 92.3

Round 3 - 5 records

Input

metric	Good1	Good2	LowQ1	LowV1	LowDSe1	GROSS
E_Se	100	100	100	100	100	100
E_+P	100	100	100	100	100	100
D_Se	95	94	90	92	55	86.3
D_+P	96	95	90	93	85	92.3
D_Spec	99	98.4	98	98.6	97.3	98.3
D_NPV	98.8	98	98	98.4	91.5	96.8

weakest metric = D_Se 86.3

Recompute D_Se if that record is dropped

drop →	Good1	Good2	LowQ1	LowV1	LowDSe1
D_Se becomes	84	83.6	85.5	85.1	93★

drop LowDSe1 — D_Se 86.3 → 93

Result — keep Good1 • Good2 • LowQ1 • LowV1

gross E_Se 100 · E_+P 100 · D_Se 93 · D_+P 93.8 · D_Spec 98.5 · D_NPV 98.3

spec-npv — Optimizes only **D_Spec** and **D_NPV**. It removes **LowSpec1**, then **LowSpec2**; once **D_Spec** improves, **D_NPV** becomes the weakest and **LowDSe1** is removed.

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Objective spec-npv (keep N = 4)

Round 1 - 7 records

Input									
metric	Good1	Good2	LowSpec1	LowSpec2	LowQ1	LowV1	LowDSe1	GROSS	
E_Se	100	100	93.3	92.9	100	100	100	95.3	
E_+P	100	100	92.9	92.3	100	100	100	95.1	
D_Se	95	94	96	95	90	92	55	88.4	
D_+P	96	95	35	34	90	93	85	65.2	
D_Spec	99	98.4	68.1	69.5	98	98.6	97.3	89.5	
D_NPV	98.8	98	99	98.9	98	98.4	91.5	97.3	

weakest metric = D_Spec 89.5

Recompute D_Spec if that record is dropped

drop -- Good1 Good2 LowSpec1 LowSpec2 LowQ1 LowV1 LowDSe1

D_Spec becomes 88 88.2 93.2* 93 88.1 88 88.2

drop LowSpec1 — D_Spec 89.5 → 93.2

Round 2 - 6 records

Input									
metric	Good1	Good2	LowSpec2	LowQ1	LowV1	LowDSe1	GROSS		
E_Se	100	100	92.9	100	100	100	96.4		
E_+P	100	100	92.3	100	100	100	96.3		
D_Se	95	94	95	90	92	55	87.4		
D_+P	96	95	34	90	93	85	74.7		
D_Spec	99	98.4	69.5	98	98.6	97.3	93.2		
D_NPV	98.8	98	98.9	98	98.4	91.5	97.1		

weakest metric = D_Spec 93.2

Recompute D_Spec if that record is dropped

drop -- Good1 Good2 LowSpec2 LowQ1 LowV1 LowDSe1

D_Spec becomes 92.1 92.3 98.3* 92.2 92.1 92.4

drop LowSpec2 — D_Spec 93.2 → 98.3

Round 3 - 5 records

Input									
metric	Good1	Good2	LowQ1	LowV1	LowDSe1	GROSS			
E_Se	100	100	100	100	100	100	100		
E_+P	100	100	100	100	100	100	100		
D_Se	95	94	90	92	55	86.3			
D_+P	96	95	90	93	85	92.3			
D_Spec	99	98.4	98	98.6	97.3	98.3			
D_NPV	98.8	98	98	98.4	91.5	96.8			

weakest metric = D_NPV 96.8

Recompute D_NPV if that record is dropped

drop -- Good1 Good2 LowQ1 LowV1 LowDSe1

D_NPV becomes 96.4 96.6 96.5 96.4 98.3*

drop LowDSe1 — D_NPV 96.8 → 98.3

Result — keep Good1 · Good2 · LowQ1 · LowV1

gross E_Se 100 · E_+P 100 · D_Se 93 · D_+P 93.8 · D_Spec 98.5 · D_NPV 98.3

sens — Optimizes only E_Se and D_Se. Since D_Se is the weakest, it removes LowDSe1 and keeps LowSpec1/2 despite their low specificity. Final D_Spec drops to 83.0.

Objective sens (keep N = 4)

Round 1 - 7 records

Input									
metric	Good1	Good2	LowSpec1	LowSpec2	LowQ1	LowV1	LowDSe1	GROSS	
E_Se	100	100	93.3	92.9	100	100	100	95.3	
E_+P	100	100	92.9	92.3	100	100	100	95.1	
D_Se	95	94	96	95	90	92	55	88.4	
D_+P	96	95	35	34	90	93	85	65.2	
D_Spec	99	98.4	68.1	69.5	98	98.6	97.3	89.5	
D_NPV	98.8	98	99	98.9	98	98.4	91.5	97.3	

weakest metric = D_Se 88.4

Recompute D_Se if that record is dropped

drop -- Good1 Good2 LowSpec1 LowSpec2 LowQ1 LowV1 LowDSe1

D_Se becomes 87.2 87 87.4 87.6 88.2 87.9 93.7*

drop LowDSe1 — D_Se 88.4 → 93.7

Round 2 - 6 records

Input									
metric	Good1	Good2	LowSpec1	LowSpec2	LowQ1	LowV1	GROSS		
E_Se	100	100	93.3	92.9	100	100	95		
E_+P	100	100	92.9	92.3	100	100	94.7		
D_Se	95	94	96	95	90	92	93.7		
D_+P	96	95	35	34	90	93	63.3		
D_Spec	99	98.4	68.1	69.5	98	98.6	88.2		
D_NPV	98.8	98	99	98.9	98	98.4	88.5		

weakest metric = D_Se 93.7

Recompute D_Se if that record is dropped

drop -- Good1 Good2 LowSpec1 LowSpec2 LowQ1 LowV1

D_Se becomes 93.3 93.6 93.3 93.5 94.3* 94

drop LowQ1 — D_Se 93.7 → 94.3

Round 3 - 5 records

Input									
metric	Good1	Good2	LowSpec1	LowSpec2	LowV1	GROSS			
E_Se	100	100	93.3	92.9	100	94.7			
E_+P	100	100	92.9	92.3	100	94.4			
D_Se	95	94	96	95	92	94.3			
D_+P	96	95	35	34	93	60.2			
D_Spec	99	98.4	68.1	69.5	98.6	86.2			
D_NPV	98.8	98	99	98.9	98.4	98.6			

weakest metric = D_Se 94.3

Recompute D_Se if that record is dropped

drop -- Good1 Good2 LowSpec1 LowSpec2 LowV1

D_Se becomes 94.1 94.5 94 94.2 94.9*

drop LowV1 — D_Se 94.3 → 94.9

Result — keep Good1 · Good2 · LowSpec1 · LowSpec2

gross E_Se 94.4 · E_+P 94.1 · D_Se 94.9 · D_+P 55.9 · D_Spec 83 · D_NPV 98.6

Ceil E_Se — Sets an upper limit for E_Se (e.g. 95.5). Greedy skips removals that would increase E_Se above the cap, allowing Spec/NPV to improve without unnecessarily increasing E_Se.

The example below uses demo_ceil.xlsx.

Ceil E_Se — round by round (cap = 96, objective spec-npv, keep 4)

Round 1 - 6 records · gross E_Se 97.1 (above cap)

Input									
metric	Good1	Good2	BadHiE1	BadHiE2	BadLoE	Mid	GROSS		
E_Se	100	100	100	100	87.5	100	97.1		
E_+P	100	100	83.3	83.3	85.7	100	91.2		
D_Se	95	94	95	95	95	90	94		
D_+P	96	95	40	41	39	90	57.1		
D_Spec	99	98.6	70.6	72.5	69.2	98	84.3		
D_NPV	98.8	98.3	98.8	98.9	98.7	98	98.5		

weakest metric = D_Spec 84.3 | ceil active (E_Se 97.1 > 96)

Recompute if that record is dropped

drop -- Good1 Good2 BadHiE1 BadHiE2 BadLoE Mid

D_Spec becomes 81.5 81.6 87.1* 86.7 87.4 81.5

E_Se becomes 96.7 96.7 96.6 96.6 100 X 96.7

BadLoE gives the biggest D_Spec lift but would raise E_Se X blocked — so drop BadHiE1 (keeps E_Se flat)

Round 2 - 5 records · gross E_Se 96.6 (above cap)

Input + GROSS									
metric	Good1	Good2	BadHiE2	BadLoE	Mid	GROSS			
E_Se	100	100	100	87.5	100	96.6			
E_+P	100	100	83.3	85.7	100	92.9			
D_Se	95	94	95	95	90	93.8			
D_+P	96	95	41	39	90	62.2			
D_Spec	99	98.6	72.5	69.2	98	87.1			
D_NPV	98.8	98.3	98.9	98.7	98	98.5			

weakest metric = D_Spec 87.1 | ceil active (E_Se 96.6 > 96)

Recompute if that record is dropped

drop -- Good1 Good2 BadHiE2 BadLoE Mid

D_Spec becomes 84.2 84.4 91* 91.7 84.4

E_Se becomes 95.8 95.8 95.7 100 X 95.8

BadLoE gives the biggest D_Spec lift but would raise E_Se X blocked — so drop BadHiE2 (keeps E_Se flat)

Result — keep Good1 · Good2 · BadLoE · Mid

gross E_Se 95.7 (held at the cap), D_Spec 91

flag modes

Flag mode removes records that fail a fixed metric threshold, then recomputes the gross metrics of the remaining records. A flagged sheet lists the removed records for review.

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Select the **Mode** and set the threshold; the greedy fields are disabled.

cut annann to deviceevent typesvalidatemetric mergefilterhr report

Inputs: metrics .xlsx files (or a folder of them)W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\filter\demo\demo_filter.xlsx

1Add files...Add folder...Remove selectedClear

2flag: low D_Se

Mode

Low duration sensitivity: DSe < thr (device caught AFIB but under-covers its duration).

Filter (flag modes)

flag-Q: Q_Se <92.79

flag-Q: Q_+P <98.87

flag-V: V_Se <87.58

flag-V: V_+P <19.09

flag-low-Dse: DSe <90.0

3Greedy select

N — records to keep300

Objectivespec-npv

Ceil E_Se (empty = off)95.5

Prefilter (drop metric <= 0)☒

Output metrics summary .xlsxW:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\filter_greedy_300.xlsxBrowse

Run

4In output

Open log

1Add files → demo_filter.xlsx

2Mode = flag: low D_Se

3set the threshold

4Run

Report table (select cells + Ctrl+C, or the buttons):

Copy table

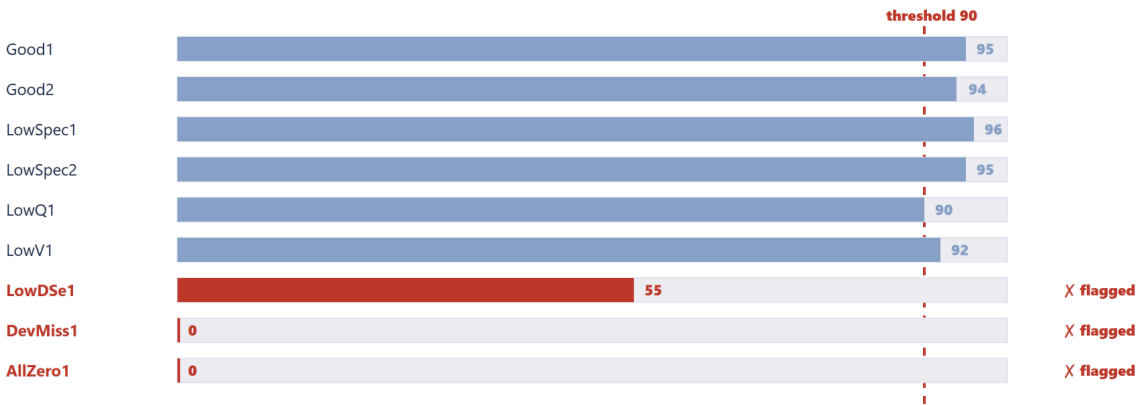
Save .csv

Five flag modes are available:

- **Low Q_Se / Q_+P** — Beat detection.
- **Low V_Se / V_+P** — Beat detection.
- **Low D_Se** — AFIB duration sensitivity.
- **Device miss/false AFIB** — AFIB detection errors.
- **AFIB all-zero** — AFIB metrics are all zero.

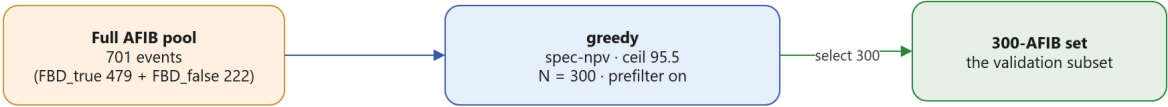
Example: **Low D_Se < 90** removes records below 90, then recalculates the remaining metrics.

Example D_Se < 90



End-to-end: reproduce the finalised 300-AFIB set

300-AFIB validation set



HR report tab

Average HR over AFIB episodes (**avg_hr = 60/mean(RR/250)**) + a **<60 / 60–120 / >120** bpm split.

Run it (markers 1-4):

- 0. **Download** the source folder **..\test-data\full-dataset\Tech\AFIB\FBD_true** (479 events) from the **full-dataset** on SharePoint (linked in the [Dataset](#) section above).

cut annann to deviceevent typesvalidatemetric mergefilterhr report

1Source folderW:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\full-dataset\Tech\AFIB\FBD_trueBrowse

2Verify (section)tech

3RunOpen outputOpen log

42026-08-12 16:16:55 =====
2026-08-12 16:16:55 === RUN @ 2026-08-12 16:16:55 ===
2026-08-12 16:16:55 source=W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\full-dataset\Tech\AFIB\FBD_true | verify=tech -> section=strip
2026-08-12 16:16:55 Total events: 479 | with avg_hr (AFIB): 479 | no AFIB: 0
2026-08-12 16:16:55 HR<60=89 | HR 60-120=294 | HR>120=96
2026-08-12 16:16:55 avg_hr: min=31.4 mean=91.1 max=216.2
2026-08-12 16:16:55 Saved -> W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\test-data\examples\hr_479.xlsx
2026-08-12 16:16:55 === DONE ===
2026-08-12 16:16:55 log appended: W:\report\HES_VALIDATION\repo-misc-tool\hes-fda-validation\hes-toolkit\logs\2026-08-12_hr_report.log (use 'Open log')

1Source folder — the event folder to score (input).

2Verify (section) — which annotation the HR is read from — ai or tech (here tech).

3Run

4Result — 479 events · avg HR min 31.4 / mean 91.1 / max 216.2 · bands <60=89, 60-120=294, >120=96.

- 1. **Source folder** — the event folder to score (input).
- 2. **Verify (section)** — which annotation the HR is read from: **ai** or **tech** (here **tech**).
- 3. **Run**.
- 4. **Result** — 479 events; avg HR min 31.4 / mean 91.1 / max 216.2 bpm; bands **<60**=89, **60–120**=294, **>120**=96.